

Prasanna Govindarajulu Sairam

(585) 485-2224 | govinda2@buffalo.edu | Buffalo, NY | [linkedin.com/in/prasanna-sairam](https://www.linkedin.com/in/prasanna-sairam) | github.com/prasannaman44

EDUCATION

University at Buffalo, School of Engineering and Applied Sciences

Expected May 2027

Bachelor of Science in Computer Science

- **Relevant Coursework:** Object-Oriented Programming & Data Structures, Algorithms & Complexity, Systems Programming, Computer Organization, Web Applications, Probability & Statistics

EXPERIENCE

Quantitative Research & AI Engineering Intern

Jun 2026 – Present

Tekly Studio

Remote

- Built an end-to-end machine learning pipeline in Python (pandas, NumPy, scikit-learn, XGBoost) to predict 7-day directional returns across ~489 cryptocurrencies (~320K daily observations), spanning feature engineering, model tuning, backtesting, and production signal generation.
- Applied rigorous financial-ML validation — purged K-fold and walk-forward cross-validation, train-only imputation, and point-in-time correctness — to prevent data leakage and look-ahead bias; validation methodology passed a line-by-line technical review.
- Diagnosed and fixed a production survivorship-bias bug in which delisted assets' stale prices surfaced as current trading signals, implementing a point-in-time universe filter; the issue was highlighted cohort-wide as a genuine production find.
- Productionized the research model into a command-line script emitting standardized JSON trading signals, and managed the full Git branch / pull request / code-review workflow, iterating on detailed reviewer feedback.

PROJECTS

Backend Web Server | *Python, MongoDB, WebSockets, bcrypt*

Sep – Dec 2025

- Built a custom HTTP server in Python from scratch, handling request parsing, routing, and response generation without high-level web frameworks.
- Designed RESTful API endpoints supporting user authentication, real-time chat messaging, and file uploads.
- Implemented secure session management with hashed cookies and bcrypt password hashing to protect user credentials.
- Integrated MongoDB to store users, chat history, and media, and added WebSocket handshakes to enable real-time messaging.

Escape from UB (Horror Survival Game) | *Unreal Engine*

Nov – Dec 2025

- Designed and built a horror survival game inspired by Five Nights at Freddy's, featuring camera surveillance, enemy AI, and player resource management.
- Implemented core gameplay systems including AI pathing, timed enemy movement, jumpscare triggers, and fail-state logic to build player tension.
- Created interactive UI elements such as camera feeds, control panels, and status indicators.

OCaml Interpreter | *OCaml*

Oct – Dec 2024

- Built a stack-based interpreter for an OCaml-like bytecode language supporting arithmetic, boolean logic, strings, control flow, and error handling.
- Implemented first-class functions and closures, including recursion, higher-order functions, and lexical environment capture.
- Added support for function calls, returns, and in/out parameters, correctly restoring stack and environment state across calls.

Memory Allocator | *C*

Nov – Dec 2024

- Developed a custom memory allocator in C, implementing malloc, free, calloc, and realloc.
- Emphasized multi-pool and bulk allocation strategies for efficient memory reuse.

Song Rater | *Java*

Feb – Mar 2024

- Built a Java application for rating songs, supporting adding, removing, and averaging ratings across multiple entries.
- Wrote comprehensive test cases covering standard and edge cases to validate functionality.

TECHNICAL SKILLS

Languages: Python, Java, C, OCaml, JavaScript, HTML/CSS

ML & Data: pandas, NumPy, scikit-learn, XGBoost, Jupyter

Tools & Technologies: Git, MongoDB, WebSockets, bcrypt, Unreal Engine, IntelliJ IDEA

Concepts: Machine Learning, Data Pipelines, Model Validation & Backtesting, REST APIs, Backend Development, Systems Programming, Memory Management, Object-Oriented Programming, Data Structures & Algorithms